

OMAR ABDELAZIZ KAPIL

Cybersecurity Specialist | Cloud Architect

Cairo, Egypt • omarkapil012@gmail.com • +20 100 977 4927 • linkedin.com/in/omar-kapil • omarkapil.github.io/my_site

PROFESSIONAL SUMMARY

Results-driven Cybersecurity Specialist and Cloud Architect with hands-on expertise in penetration testing, red teaming, security orchestration, and AWS cloud infrastructure design. Proven ability to engineer multi-layered defense systems, automate vulnerability management pipelines, and deliver enterprise-grade security solutions for SMEs. Currently completing a Bachelor of Science in Cybersecurity (Expected 2026) and seeking a challenging role to apply offensive and defensive security skills at scale.

PROFESSIONAL EXPERIENCE

Cloud Architect Intern | [National Telecommunication Institute \(NTI\)](#) Summer 2025

Cairo, Egypt · 120-Hour Intensive Program · Final Score: 87.5%

- Engineered high-availability AWS architectures including VPC configuration, EC2 instance management, S3 storage optimization, and Route 53 DNS management, enabling scalable and fault-tolerant cloud infrastructure.
- Implemented cloud security best practices encompassing IAM Roles/Policies, Security Groups, and Network ACLs to enforce least-privilege access and ensure data integrity across distributed environments.
- Architected cost-optimized infrastructure blueprints for high-availability systems, balancing performance requirements with cloud spend — achieving a final evaluation score of 87.5% out of 100.
- Collaborated cross-functionally on cloud migration strategies, applying hands-on knowledge of EBS, RDS, and subnetting to design resilient network topologies.

Security Operations & Engineering | [Helwan International Technology University \(HITU\)](#) 2022 – Present

Academic & Applied Security Research · Cairo, Egypt

- Spearheaded the design and implementation of multi-layered cybersecurity defense frameworks to protect corporate networks, hardware, and digital assets against advanced persistent threats (APTs) and cyber-attacks.
- Engineered Intrusion Detection and Prevention Systems (IDS/IPS) using advanced behavioral modeling, reducing mean time-to-detect (MTTD) threats in virtualized environments.
- Executed penetration testing and ethical hacking engagements using Metasploit, Nmap, and Wireshark, identifying and remediating critical vulnerabilities across target systems.
- Conducted cybersecurity auditing and compliance assessments aligned with international security frameworks, improving organizational security posture scores.
- Deployed applied cryptographic solutions utilizing symmetric (AES) and asymmetric (RSA) encryption algorithms to ensure data confidentiality, integrity, and non-repudiation across communication platforms.
- Administered Linux system hardening (Ubuntu, Parrot OS, RHEL), implementing kernel-level security configurations and reducing the attack surface of critical infrastructure.
- Designed and implemented distributed Database Management Systems (DBMS) using advanced SQL, table normalization, and object-oriented database structures to secure sensitive organizational data.

KEY PROJECTS

Automated Security Orchestration Platform For SMEs — *Project Lead — 11-Member Team*

- Architected a comprehensive security dashboard delivering enterprise-grade vulnerability management to SMEs, automating the full discovery-to-report pipeline using Python/Flask, Docker, Nmap API, and OpenVAS API.
- Reduced manual workload for security audits by building an automated orchestration engine that eliminates repetitive scanning and reporting tasks for security teams.
- Led a cross-functional team of 11 engineers through all SDLC phases — requirements, design, development, testing, and deployment — delivering the platform on schedule.
- Integrated containerized microservices using Docker to ensure scalable, reproducible deployments across diverse SME environments.

Network Intrusion Detection System (IDS) — *Developer*

- Built an intelligent ML-powered monitoring system in Python (Scikit-learn) capable of distinguishing normal network traffic from malicious activity, including DDoS and brute force attacks in real-time.
- Optimized the model's detection threshold to minimize false positives, enabling security teams to focus resources exclusively on genuine threat events.

ML-Powered Phishing Email Detector — *Developer*

- Engineered a proactive NLP-based defense tool that analyzes email headers, body text, and URL structures to classify and flag phishing attempts before user interaction.
- Automated the first line of defense against social engineering attacks and credential theft.
- Applied Natural Language Processing (NLP) pipelines to extract features from unstructured email data, enabling detection of sophisticated spear-phishing campaigns.

TECHNICAL SKILLS

Offensive Security: Penetration Testing, Ethical Hacking, Vulnerability Assessment, Red Teaming

Defensive Security: IDS/IPS Design, Multi-Layered Defense Frameworks, Security Orchestration, Incident Response

Cloud Platforms: Amazon Web Services (AWS): EC2, S3, VPC, IAM, RDS, EBS, Route 53, Security Groups, Network ACLs

Cloud Architecture: High Availability (HA), Fault Tolerance, Scalability, Cost Optimization, Cloud Migration

Security Tools: Nmap, Wireshark, Metasploit, OpenVAS

Programming Languages: Python, Java, SQL, Bash/Shell Scripting

Frameworks: Python, Docker

Cryptography: Symmetric Encryption (AES), Asymmetric Encryption (RSA), Data Integrity, Non-Repudiation

Operating Systems: Linux (Ubuntu, Kali), Windows, Windows Server

Compliance & Governance: Cybersecurity Auditing, Risk Management, IAM, Cybersecurity Law

Databases: SQL, Distributed DBMS, Object-Oriented Databases, Data Normalization

Soft Skills: Technical Documentation, Cross-Functional Team Leadership, Project Management, Analytical Problem Solving

EDUCATION

Bachelor of Science in Cybersecurity | [Helwan International Technology University \(HITU\)](#) 2022 – 2026
(Expected)

Senior Year · Cairo, Egypt

- Relevant Coursework: Penetration Testing & Ethical Hacking, Network Security, Applied Cryptography, Cloud Computing, Intrusion Detection Systems, Database Security, Artificial Intelligence, Big Data Analytics, Cybersecurity Law & Compliance, Software Development Lifecycle (SDLC), Linux System Administration.
- Directed graduation project leading a 10-member cross-functional team through the comprehensive research, design, and implementation phases of an enterprise-grade security orchestration platform.

CERTIFICATIONS & TRAINING

- AWS Cloud Architecture — National Telecommunication Institute (NTI), 120-Hour Program, Score: 87.5% (Summer 2025)
- [CCNA, CompTIA Security+, CEH, eJPT, AWS Certified Solutions Architect — Qualified for it]

LANGUAGES

Arabic: Native

English: Professional Working Proficiency